Solving Linear Programs using ZIMPL and SCIP

Lê Xuân Thanh

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2. Introduction to ZIMPL and SCIP
   - What and Why
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The need of having LP solvers

- Linear programs in real-life often have large sizes
- Human computational performance is limited
- Era of industry 4.0

$\Rightarrow$ The need of computational softwares

Example: Diet problem of Stigler[2]

Posed by Stigler in 1945[1]:

> *For a moderately active man (economist) weighing 154 pounds, how much of each of 77 foods should be eaten on a daily basis so that the man's intake of nine nutrients (including calories) will be at least equal to the recommended dietary allowances (RDAs) suggested by the National Research Council in 1943, with the cost of the diet being minimal?*

---

[1] G. J. Stigler. The cost of subsistence. *Journal of Farm Economics*, 27(2):303–314, 1945

[2] George Joseph Stigler (17.01.1911-01.12.1991): an American economist, Nobel Memorial Prize in Economic Sciences 1982

Example: Diet problem of Stigler (cont.)

TABLE 1. DAILY ALLOWANCES OF NUTRIENTS FOR A MODERATELY ACTIVE MAN (weighing 154 pounds)*

Nutrient	Allowance
Calories	3,000 calories
Protein	70 grams
Calcium	.8 grams
Iron	12 milligrams
Vitamin A	5,000 International Units
Thiamine (B_1)	1.8 milligrams
Riboflavin (B_2 or G)	2.7 milligrams
Niacin (Nicotinic Acid)	18 milligrams
Ascorbic Acid (C)	75 milligrams

* National Research Council, *Recommended Dietary Allowances*, Reprint and Circular Series No. 115, January, 1943.

Example: Diet problem of Stigler (cont.)

Nutrients and prices of 77 foods (in USD, year 1945)

`https://github.com/lxthanh86/StiglerDiet`

Solution of Stigler

TABLE 2. MINIMUM COST ANNUAL DIETS, AUGUST 1939 AND 1944

Commodity	August 1939		August 1944	
	Quantity	Cost	Quantity	Cost
Wheat Flour	370 lb.	$13.33	535 lb.	$34.53
Evaporated Milk	57 cans	3.84	—	—
Cabbage	111 lb.	4.11	107 lb.	5.23
Spinach	23 lb.	1.85	13 lb.	1.56
Dried Navy Beans	285 lb.	16.80	—	—
Pancake Flour	—	—	134 lb.	13.08
Pork Liver	—	—	25 lb.	5.48
Total Cost		$39.93		$59.88

Example: Diet problem of Stigler (cont.)

- Stigler did not use simplex method
- Need 9 accountants working in 120 days to solve by hand using Stigler's method
- By SCIP: 0.01 second!

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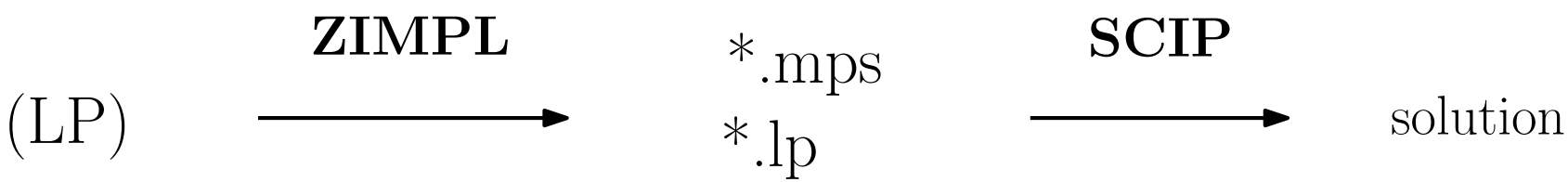
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From real-life problems to their solutions

Many real-life problems can be formulated as linear programs.



What is ZIMPL?

(LP) $\xrightarrow{\textbf{ZIMPL}}$ *.mps, *.lp



Prof. Thorsten Koch

- ZIMPL[3]:
  - Zuse Institut Mathematical Programming Language
  - https://zimpl.zib.de/

[3] Thorsten Koch. *Rapid Mathematical Programming.* PhD thesis, Technische Universität Berlin, 2004

Alternatives for ZIMPL

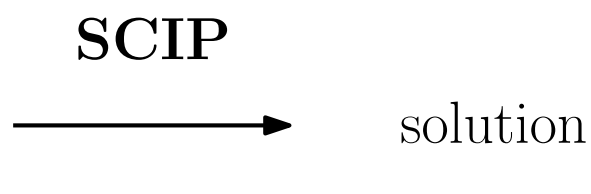
Name	**Full name**	**URL**
AIMMS	Advanced Integrated Multi-dimensional Modeling Software	www.aimms.com
AMPL	A Modeling Language for Mathematical Programming	www.ampl.com
GAMS	General Algebraic Modeling System	www.gams.com
LINGO	Lingo	www.lindo.com
LPL	(Linear|Logical|Literate) Programming Language	www.virtual-optima.com
MINOPT	Mixed Integer Non-linear Optimizer	titan.princeton.edu/MINOPT
MOSEL	Mosel	www.dashoptimization.com
MPL	Mathematical Programming Language	www.maximalsoftware.com
OMNI	Omni	www.haverly.com
OPL	Optimization Programming Language	www.ilog.com
GNU-MP	GNU Mathematical Programming Language	www.gnu.org/software/glpk

Why ZIMPL?

Special features of ZIMPL:

- Free, open source
- Independent of solver
- Infinite rational
- Easy to install, easy to use

What is SCIP?





Dr. Tobias Achterberg

- SCIP[4]:
  - Solving Constraint Integer Programs
  - https://scip.zib.de/

[4] Tobias Achterberg. *Constraint Integer Programming.* PhD thesis, Technische Universität Berlin, 2007

Alternatives for SCIP

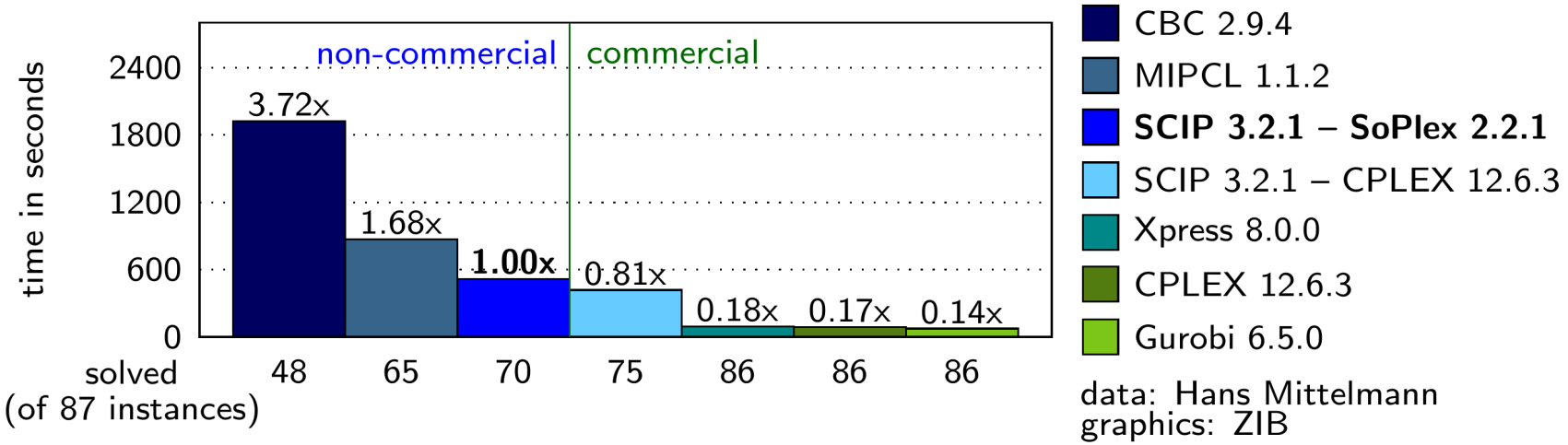
Name	**Full name**	**URL**
CBC	COIN Branch and Cut	www.coin-or.org/Cbc/
GLPL	GNU Linear Programming Kit	www.gnu.org/software/glpk/
MIPCL	Mixed Integer Programming Class Library	
SoPlex	Sequential object-oriented simPlex	https://soplex.zib.de/
Xpress	FICO Xpress optimizer	www.fico.com/en/products/fico-xpress-solver
CPLEX	IBM CPLEX Optimizer	www.ibm.com/analytics/cplex-optimizer
GUROBI	GUROBI	www.gurobi.com/

Why SCIP?

Special features of SCIP:

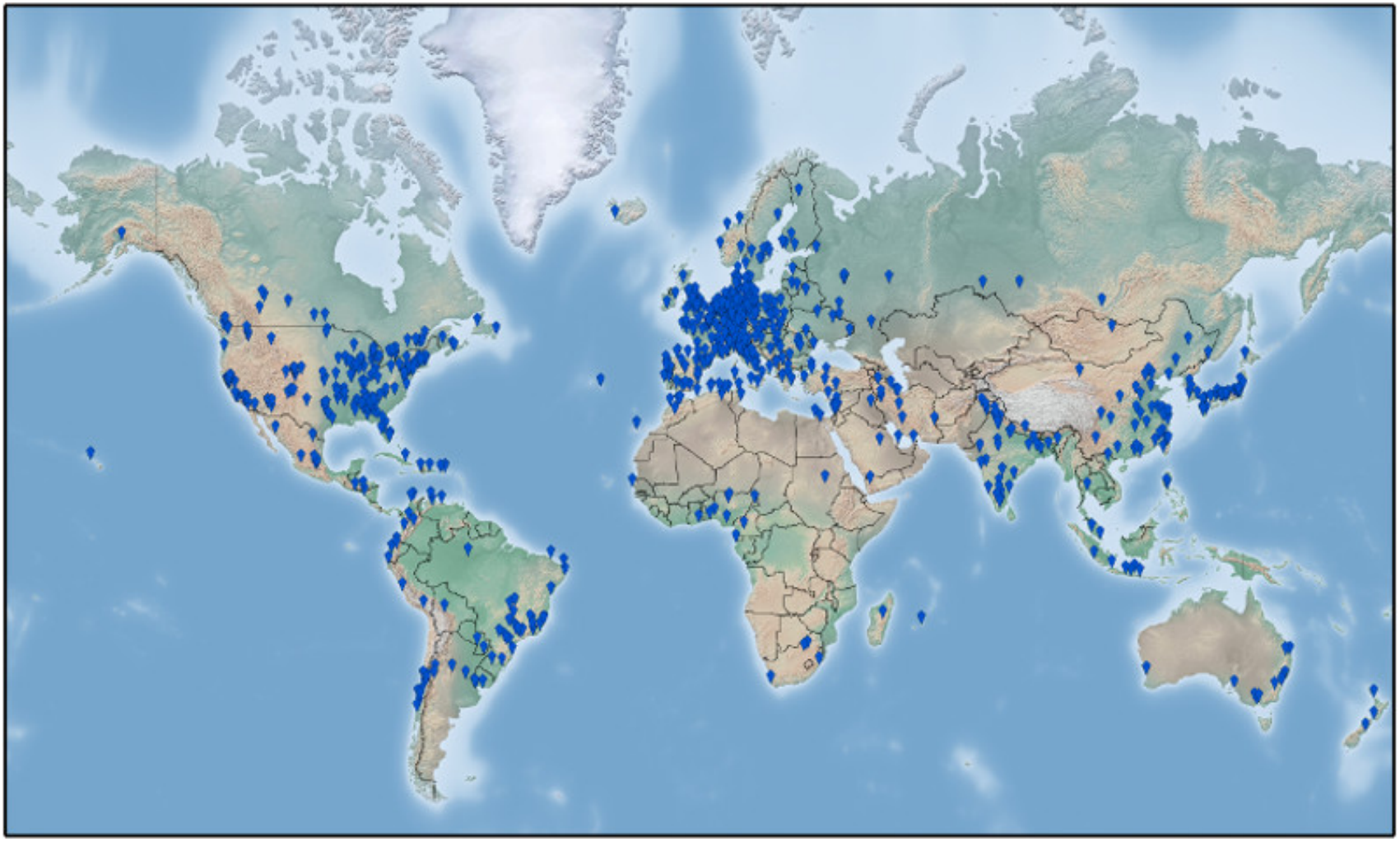
- Free, open source
- Fastest among free solvers
- Easy to install, easy to use

Why SCIP?



https://scip.zib.de/

Why SCIP?



Locations of registered institutes
(https://www.scipopt.org/)

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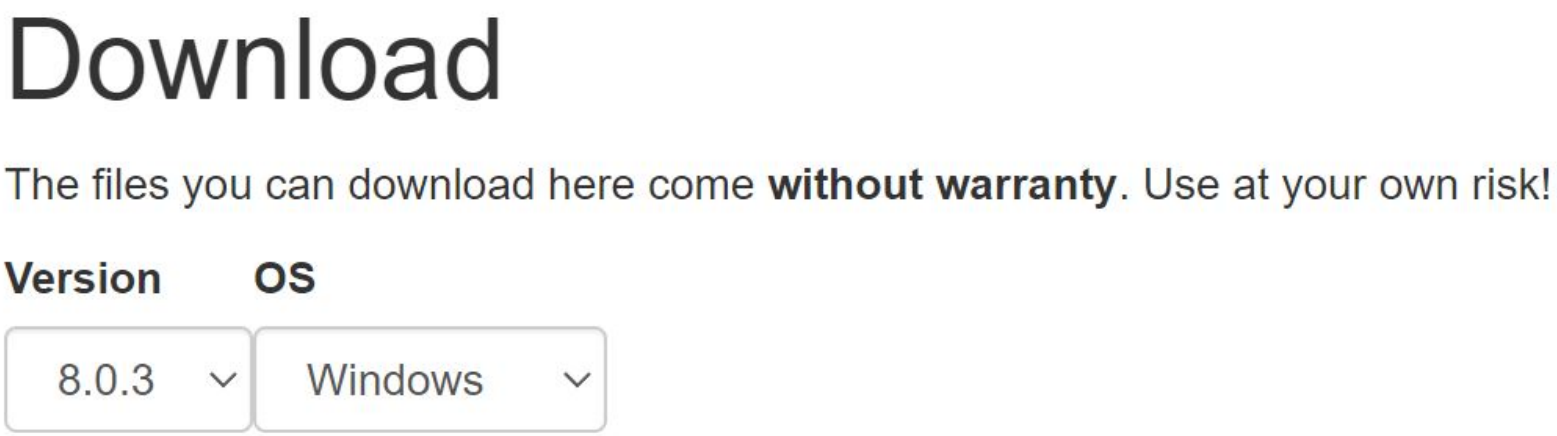
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Download ZIMPL and SCIP

Step 1: Click

`https://www.scipopt.org/index.php#download`

then choose Version 8.0.3 (latest version) and OS Windows



Download ZIMPL and SCIP (cont.)

Step 2: Go to **Precompiled Packages** and choose

- SCIPOptSuite-8.0.3-win64-VS15.exe
  (for Windows 64 bits)
- SCIPOptSuite-8.0.3-win32-VS15.exe
  (for Windows 32 bits)

Precompiled Packages

You can also download precompiled executables of SCIP with which you can solve MIP, MIQCP, CIP, SAT, or PBO instances in MPS, LP, RLP, ZIMPL, flatzinc, CNF, OPB, WBO, PIP, or CIP format.
Note that these executables do not include the readline features (i.e., command line editing and history) due to license issues. However, you can download the free readline wrapper rlwrap to provide this missing feature to the executables.

Installers (install the scipoptsuite in your computer, without source files):

	SoPlex	ZIMPL	GCG	UG	PaPILO	Ipopt	CPPAD	zlib	bliss	MPIR
SCIPOptSuite-8.0.3-win64-VS15.exe Windows 64bit installer (MSVS15); requires the Visual C++ Redistributable Packages	6.0.3	3.5.3			2.1.2	3.12.9	20180000.0		0.77	3.0.0
SCIPOptSuite-8.0.3-win32-VS15.exe Windows 64bit installer (MSVS15); requires the Visual C++ Redistributable Packages	6.0.3	3.5.3			2.1.2	3.12.9	20180000.0		0.77	3.0.0

Download ZIMPL and SCIP (cont.)

Step 3: Register before downloading

Download

File name:	SCIPOptSuite-8.0.3-win64-VS15.exe
File size:	47949 kb
MD5 sum:	35fc8fd940295cf5e221b0fc5ecc9ece
Downloads:	0

Releases up to and including Version 8.0.2 remain under the ZIB Academic License as indicated by the files contained in the releases. The Apache 2.0 License applies from Version 8.0.3 onwards.

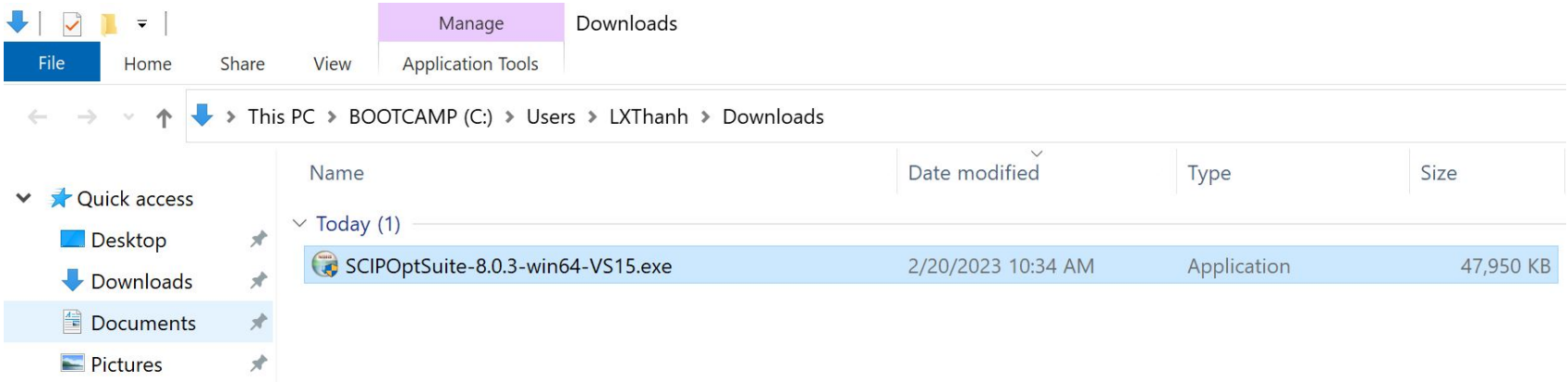
Please enter your information in the form below. Mandatory fields are marked with an (*) asterisk. The input data are only used for internal evaluation purposes.

Name*	Le Xuan Thanh
Email*	lxthanh@math.ac.vn
Institution	Institution
City	City
Country*	Vietnam

Start Download

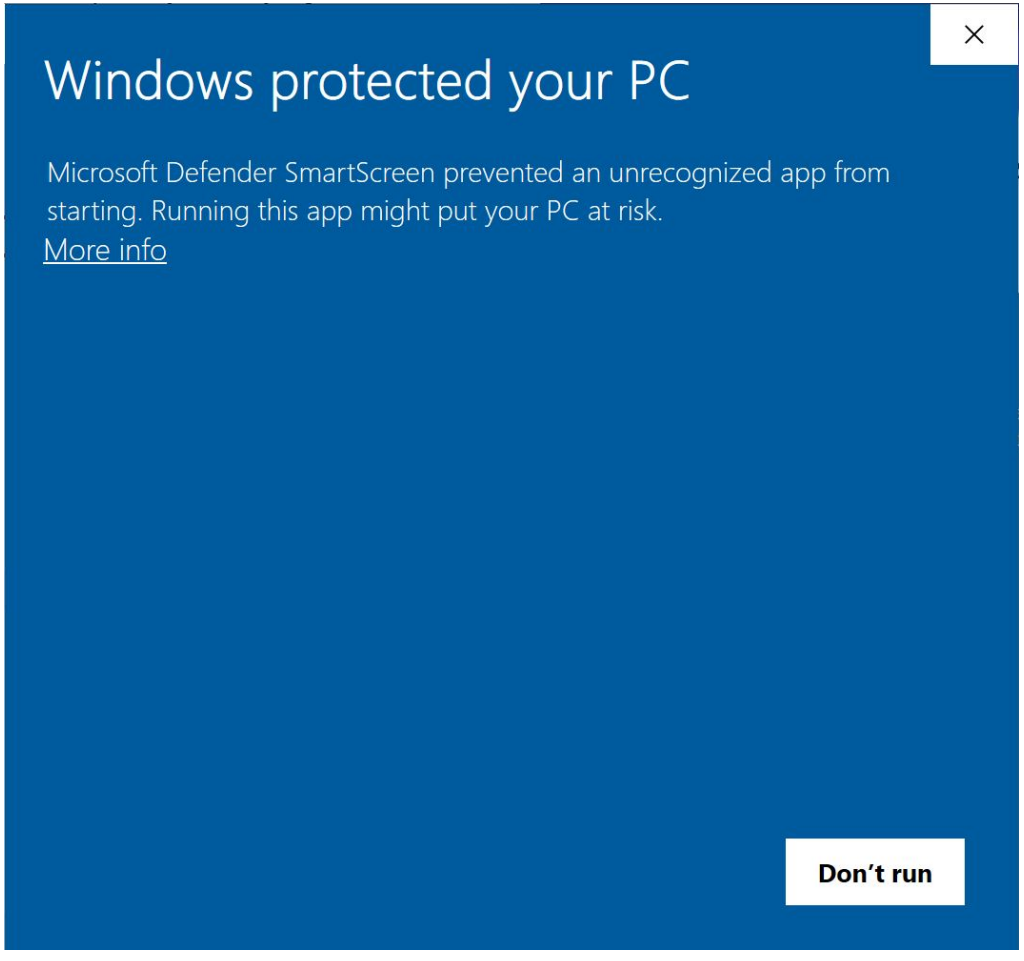
Install ZIMPL and SCIP

Step 4: Find the downloaded file and double click



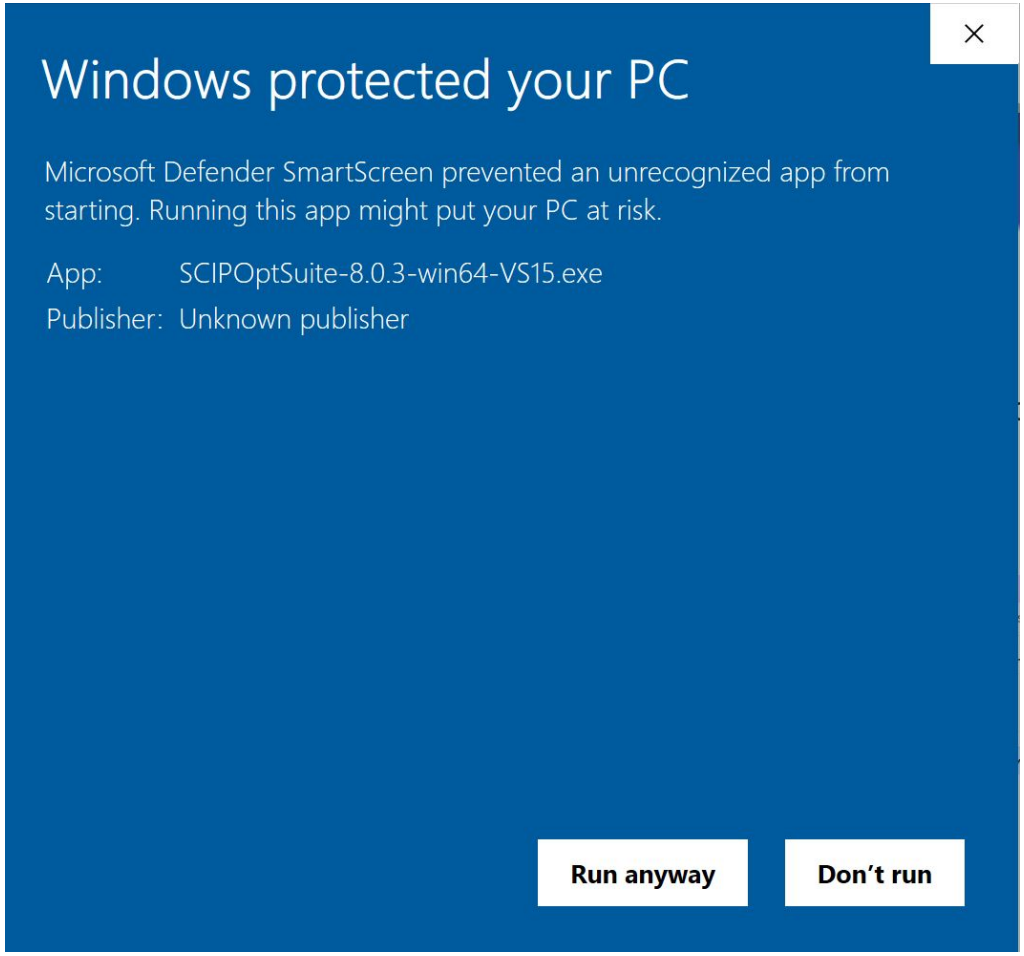
Install ZIMPL and SCIP (cont.)

Step 5: In case of unrecognized app:
choose More info then Run anyway
and follow default instruction



Install ZIMPL and SCIP (cont.)

Step 5: In case of unrecognized app:
choose More info then Run anyway
and follow default instruction

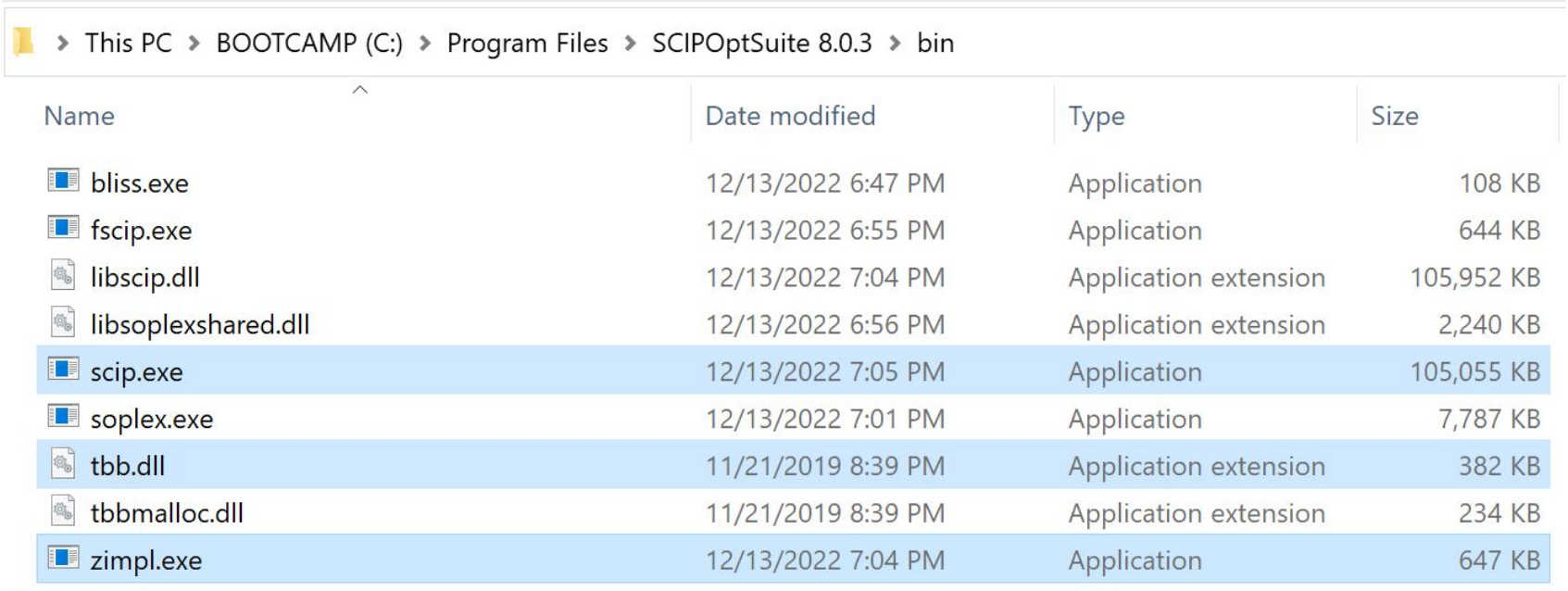


Install ZIMPL and SCIP (cont.)

Step 6: Go to

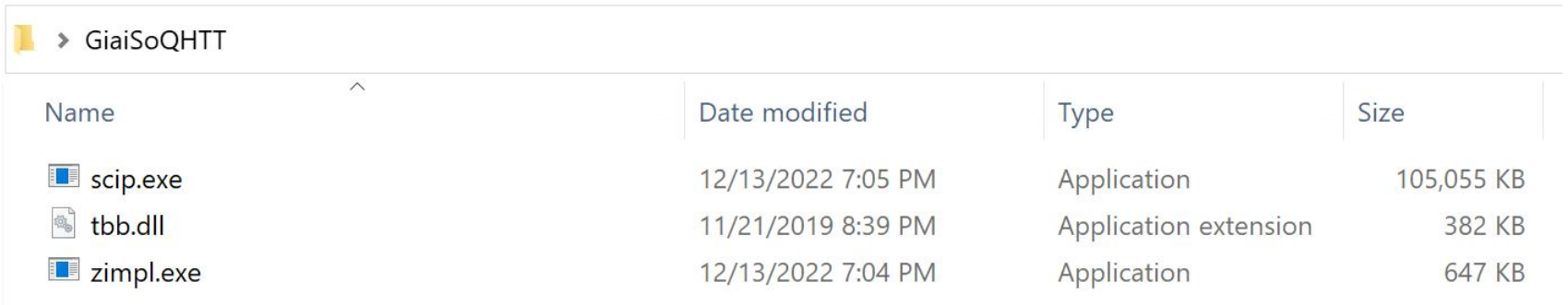
```
C:\Program Files\SCIPOptSuite 8.0.3\bin
```

and copy 3 files: scip.exe, tbb.dll, zimpl.exe



Install ZIMPL and SCIP (cont.)

Step 7: Create a folder (should be on Desktop)
and paste 3 files: scip.exe, tbb.dll, zimpl.exe into this folder



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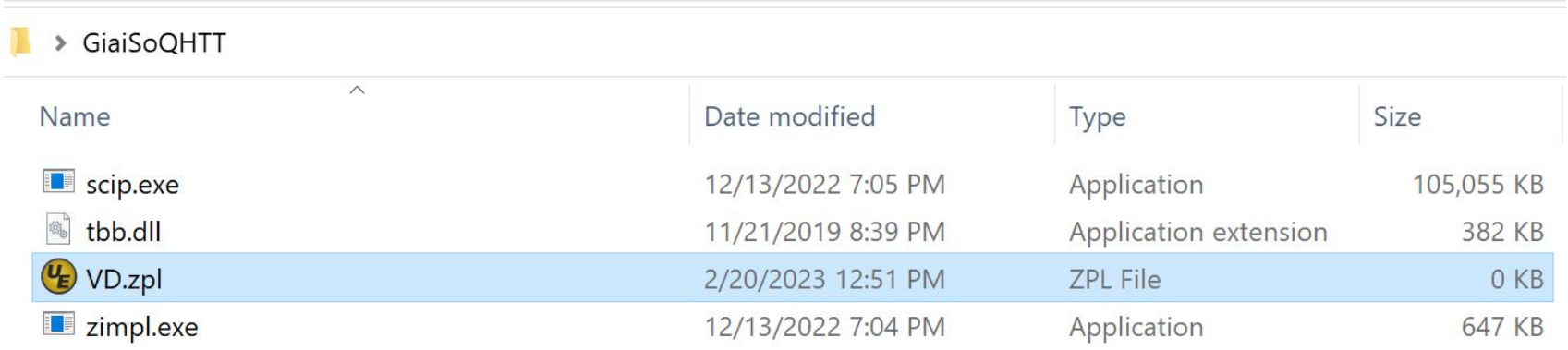
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Procedure

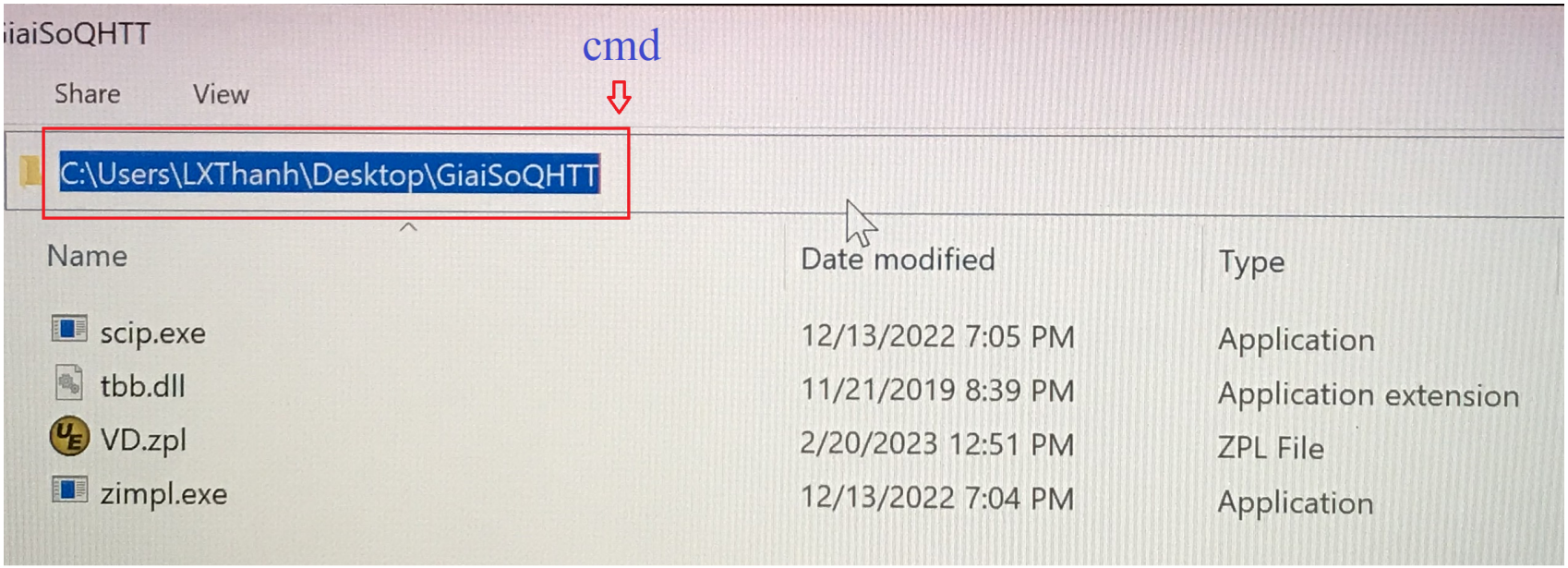
- **Step 1:**
  In the folder containing 3 files scip.exe, tbb.dll, zimpl.exe, create a text file *.zpl in which * is the file name



- **Step 2:**
  In file *.zpl, write down the ZIMPL code for the linear program to be solved
  (see examples later)

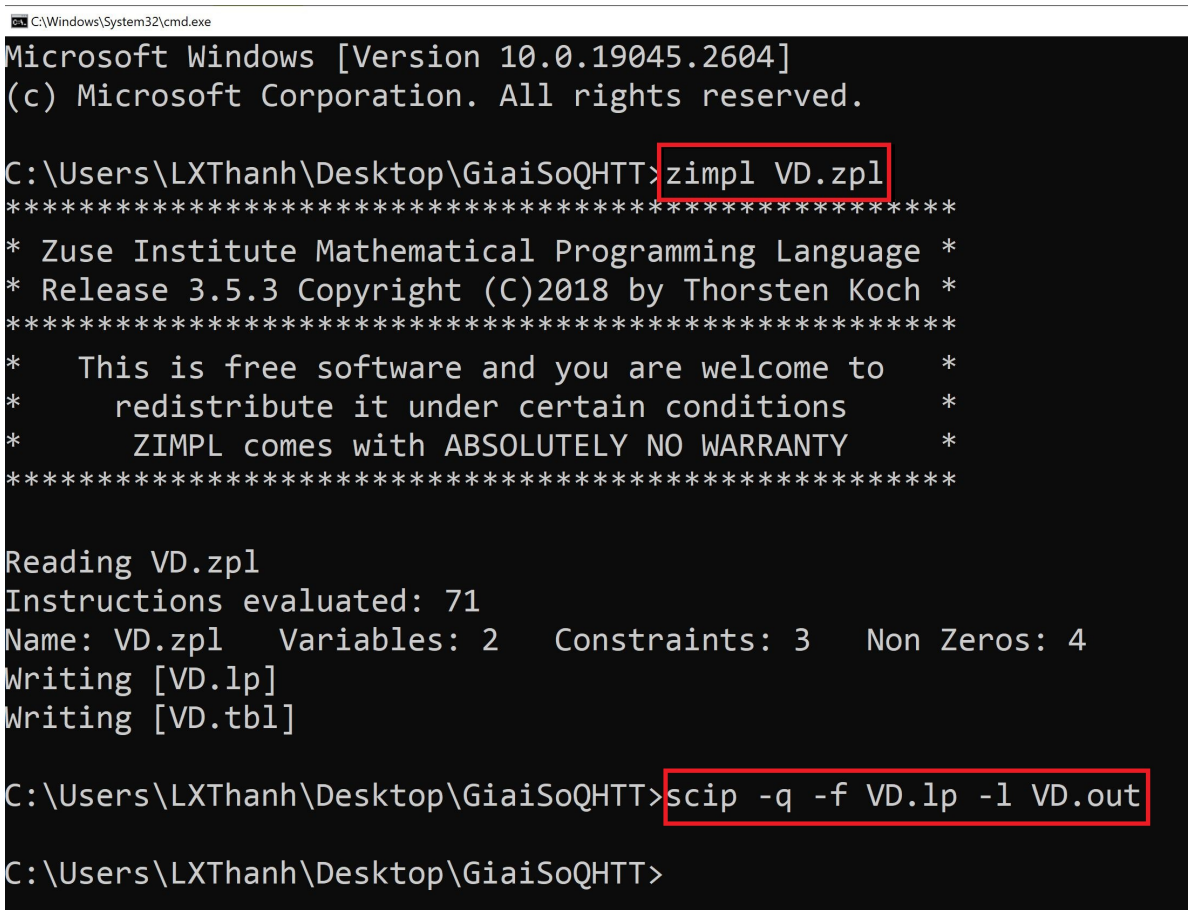
Procedure (cont.)

- **Step 3:**
  Go to the title box of the folder, type cmd then hit Enter (to open Command Prompt terminal)



Procedure (cont.)

- **Step 4:**
  In Command Prompt terminal, type the commands
  zimpl *.zpl then scip -q -f *.lp -l *.out
  in which * is the file name



Procedure (cont.)

- **Step 5:**
  Find solution in file *.out in the same folder

VD.out

```
34
35 SCIP Status        : problem is solved [optimal solution found]
36 Solving Time (sec) : 0.00
37 Solving Nodes      : 0
38 Primal Bound       : +1.9200000000000e+03 (2 solutions)
39 Dual Bound         : +1.9200000000000e+03
40 Gap                : 0.00 %
41
42 primal solution (original space):
43 ================================
44
45 objective value:                                1920
46 x1                                                30   (obj:50)
47 x2                                                10   (obj:42)
```

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Example 1

Solve

$$
\begin{aligned}
\max \quad & 50x_1 + 42x_2 \\
\text{s.t.} \quad & x_1 + x_2 \leq 40 \\
& x_1 \leq 30 \\
& x_2 \leq 30 \\
& x_1 \geq 0 \\
& x_2 \geq 0
\end{aligned}
$$

ZIMPL code

Program:

$$\begin{aligned} \max \quad & 50x_1 + 42x_2 \\ \text{subject to} \quad & \\ & x_1 + x_2 \leq 40 \\ & x_1 \leq 30 \\ & x_2 \leq 30 \\ & x_1 \geq 0 \\ & x_2 \geq 0 \end{aligned}$$

ZIMPL code:

```
# Variables
var x1 >= 0;
var x2 >= 0;

# Objective
maximize profit:
50 * x1 + 42 * x2;

# Constraints

subto Constraint1:
  x1 + x2 <= 40;

subto Constraint2:
  x1 <= 30;

subto Constraint3:
  x2 <= 30;
```

Structure of ZIMPL code

Code:

```
# Variables
var x1 >= 0;
var x2 >= 0;

# Objective
maximize profit:
50 * x1 + 42 * x2;

# Constraints

subto Constraint1:
  x1 + x2 <= 40;

subto Constraint2:
  x1 <= 30;

subto Constraint3:
  x2 <= 30;
```

Structure:

- Comment lines start with #
- Declare variables
  var [Declare variables];
- Objective:
  maximize [Name]: [Objective];
  or
  minimize [Name]: [Objective];
- Constrains:
  subto [Name]: [Constraints];

Remark:
Names of variables, objective, constraints start with alphabet characters

Results obtained by SCIP

VD.out

```
34
35 SCIP Status        : problem is solved [optimal solution found]
36 Solving Time (sec) : 0.00
37 Solving Nodes      : 0
38 Primal Bound       : +1.9200000000000e+03 (2 solutions)
39 Dual Bound         : +1.9200000000000e+03
40 Gap                : 0.00 %
41
42 primal solution (original space):
43 ================================
44
45 objective value:                                1920
46 x1                                               30   (obj:50)
47 x2                                               10   (obj:42)
```

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Example 2

Solve the following LP by using ZIMPL and SCIP:

$$
\begin{array}{lrcl}
\max & 5x_1 + 4x_2 + 3x_3 & & \\
\text{s.t.} & 2x_1 + 3x_2 + x_3 & \leq & 5 \\
& 4x_1 + x_2 + 2x_3 & \leq & 11 \\
& 3x_1 + 4x_2 + 2x_3 & \leq & 8 \\
& x_1, x_2, x_3 & \geq & 0
\end{array}
$$

Example 3

Solve the following LP by using ZIMPL and SCIP:

$$
\begin{aligned}
\max \quad & 10x_1 + 12(x_2 + y_2) + 17(x_3 + y_3) + 8(x_4 + y_4) \\
\text{s.t.} \quad & 10x_1 + 13x_2 + 15x_3 + 9x_4 \leq 1995 \\
& 25x_1 + 20y_2 + 27y_3 + 33y_4 \leq 1953 \\
& 0.1x_1 + 0.15(x_2 + y_2) + 0.5(x_3 + y_3) + 0.05(x_4 + y_4) \leq 50 \\
& x_2 + y_2 = 2(x_3 + y_3) \\
& x_1, x_2, x_3, x_4 \geq 0 \\
& y_2, y_3, y_4 \geq 0
\end{aligned}
$$

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[Recall] Stigler's diet problem

> *For a moderately active man (economist) weighing 154 pounds, how much of each of 77 foods should be eaten on a daily basis so that the man's intake of nine nutrients (including calories) will be at least equal to the recommended dietary allowances (RDAs) suggested by the National Research Council in 1943, with the cost of the diet being minimal?*

Data:

- Lists of 77 foods and 9 nutrients
- RDAs of 9 nutrients
- Amounts of nutrients in each of 77 foods
- Price of 77 foods (in USD, year 1945)

See [S1945][5] or

`https://github.com/lxthanh86/StiglerDiet`

[5] G. J. Stigler. The cost of subsistence. *Journal of Farm Economics*, 27(2):303–314, 1945

Model

Data:

- F: set of 77 foods (file Foods.dat)
- N: set of 9 nutrients (file Nutrients.dat)
- a_n: RDA of nutrient $n \in N$ per day (file Nutrients.dat)
- d_{fn}: amount of nutrient $n \in N$ in each unit of food $f \in F$ (file NutritiveValues.dat, in USD)

Variables:

x_f = payment (in USD) to buy food $f \in F$ in the diet

Model:

$$\min \quad \sum_{f \in F} x_f$$

$$\text{s.t.} \quad \sum_{f \in F} d_{fn} x_f \geq a_n \qquad \forall n \in N$$

$$x_f \geq 0 \qquad \forall f \in F$$

ZIMPL code

To show in Control Prompt terminal.

Result obtained by SCIP

```
solution status: optimal solution found
objective value:                0.10866227820675 7
x$WheatFlour(Enriched)          0.0295190616764883   (obj:1)
x$Liver(Beef)                   0.0018925572907052 7  (obj:1)
x$Cabbage                       0.0112144352461449   (obj:1)
x$Spinach                       0.0050076604667252   (obj:1)
x$NavyBeansDried                0.0610285635266932   (obj:1)
```

Thank you for your attention!

